

Drive for better vision



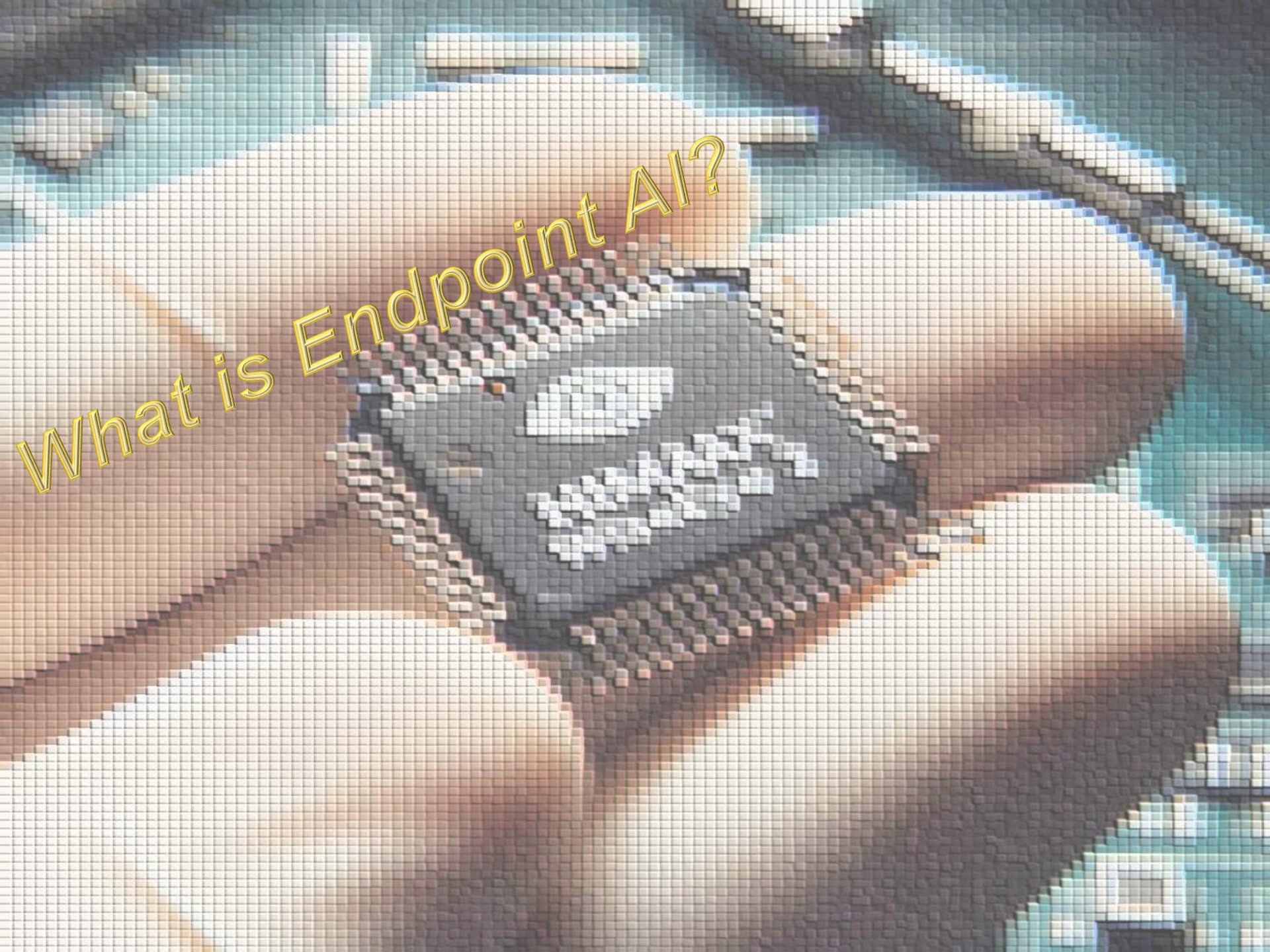
WiseEye™ 方案技術研討會

June 27, 2025

Himax Technologies, Inc.
奇景光電股份有限公司

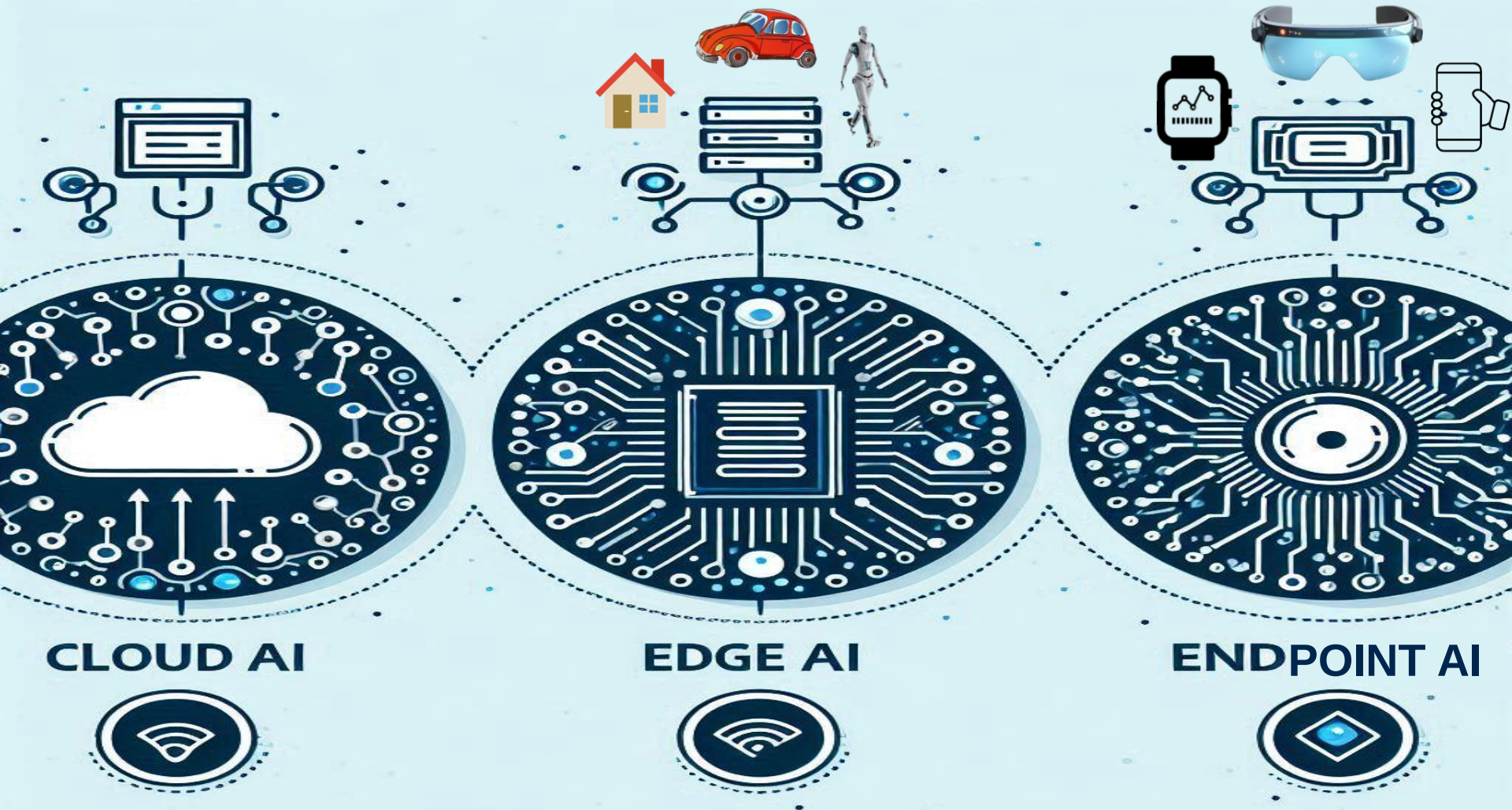
Outline

- What is Endpoint AI
- Hardware Overview
- Development Flow
- AI Model Optimization
- No Code AI platform
- Install tool at Windows environment



What is Endpoint AI?

What is Endpoint AI?

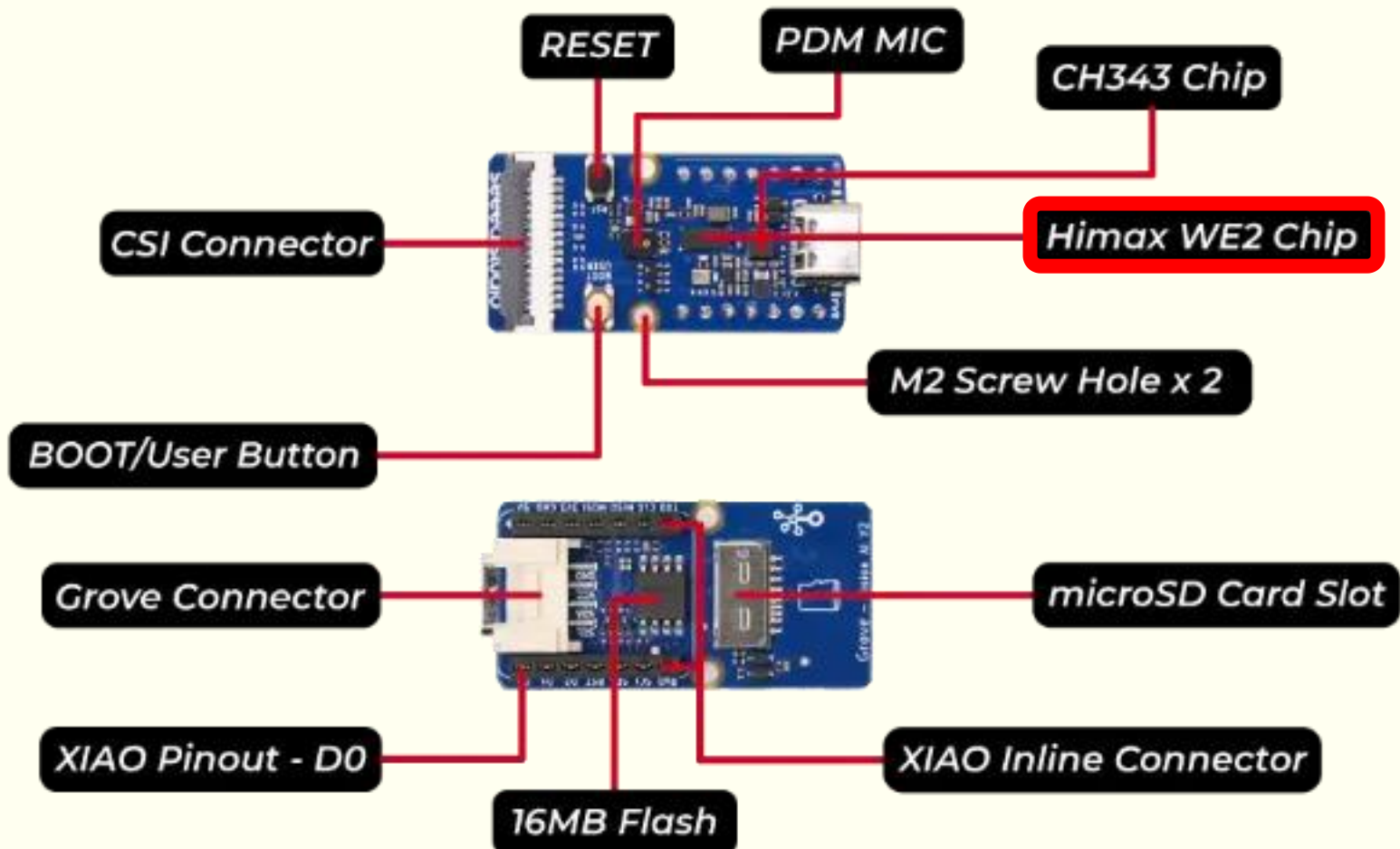


Hardware Overview



Hardware Overview

- Himax WiseEye 2 (WE2) board kit



WE2 AI Processor Block Diagram

System Control

Secure SWD

OSC & PLL

Clock Generator

Reset Generator

Timers

Watch Dog

RTC

DMA

IOMUX

Security

Crypto / TRNG

PUF

OTP

Internal Memory

64 KB Boot ROM

2 MB System SRAM

Processors

Cortex-M55 (Big)

16 KB I-cache

16 KB D-cache

256 KB I-TCM

256 KB D-TCM

Helium

FPU

TrustZone

ETM

Cortex-M55 (Little)

16 KB I-cache

16 KB D-cache

Helium

FPU

TrustZone

ETM

Ethos-U55
microNPU

Peripherals

MIPI CSI-2 RX / TX

DVP / SDI

SPI / Quad SPI

I2C / I3C

UART

PWM

GPIO

SD / SDIO

I2S TX / RX

PDM RX / TX

Sensors

ADC (4-ch)

Temp Monitor

Hardware Accelerators

Motion Detection

JPEG CODEC

HWVAD

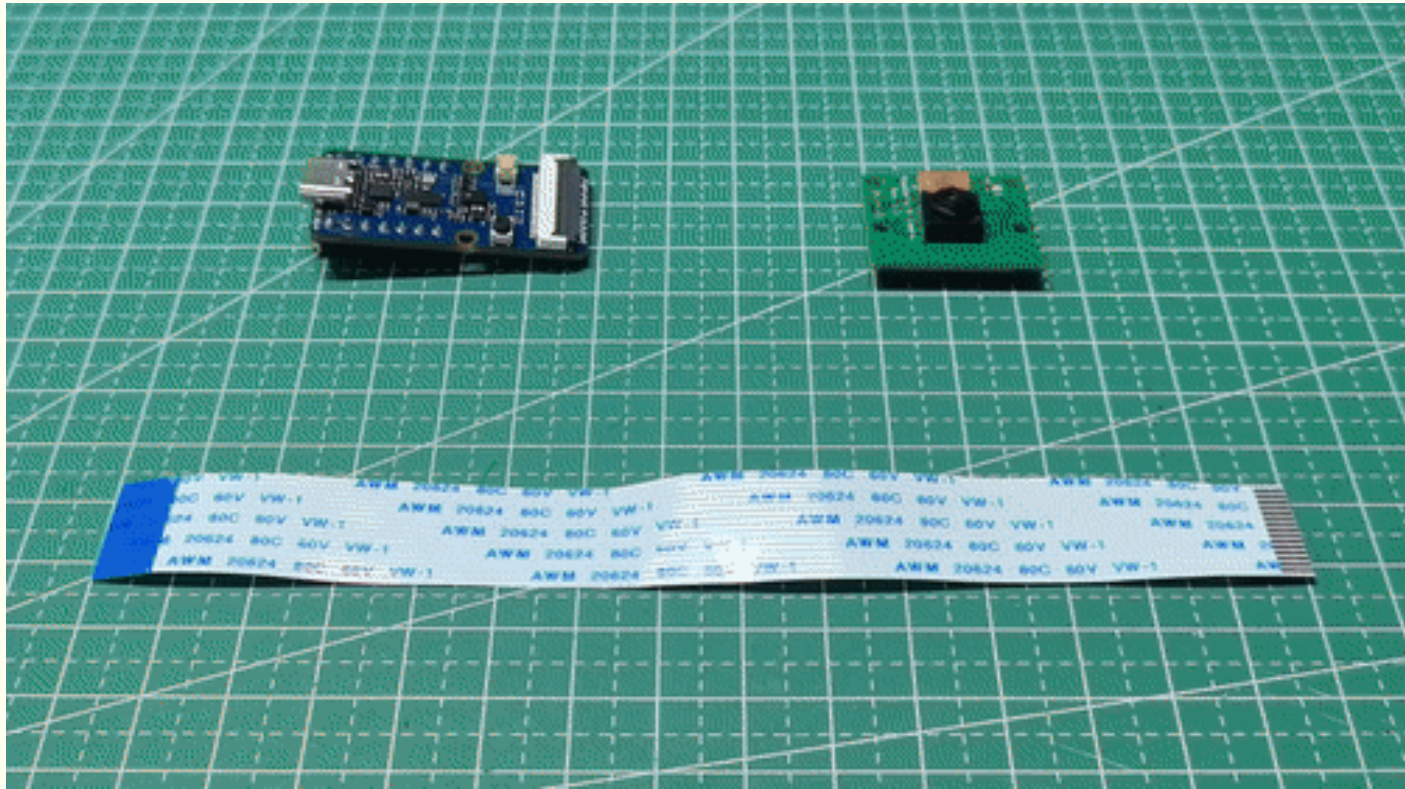
Power Management

DC-DC

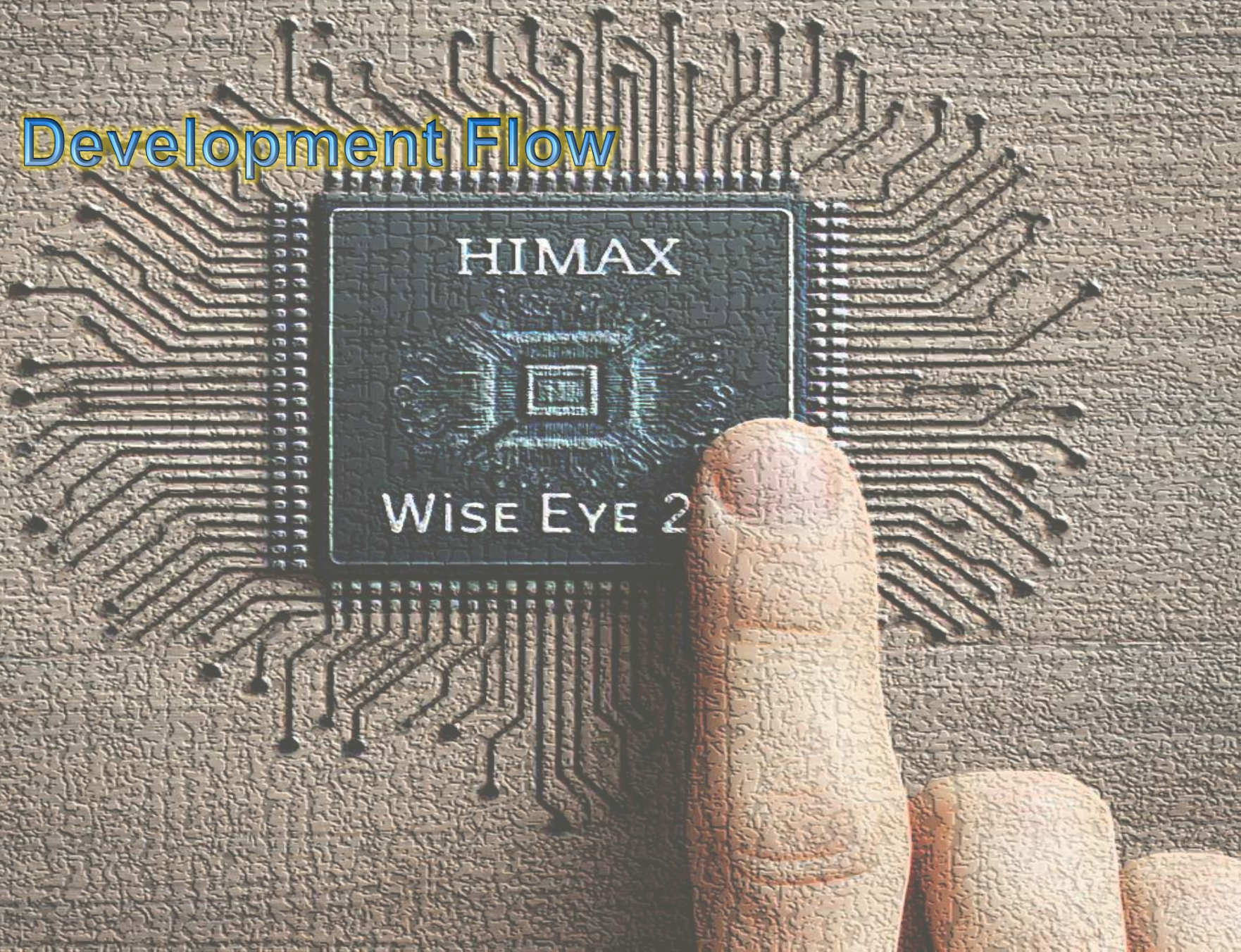
Power Management Unit

Hardware Overview

- Himax WiseEye 2 (WE2) board kit connect with camera sensor



Development Flow



Global Development Flow

 TensorFlow  PyTorch

onnx2tf tool:

<https://github.com/PINTO0309/onnx2tf>

TinyNN:

<https://github.com/alibaba/TinyNeuralNetwork>

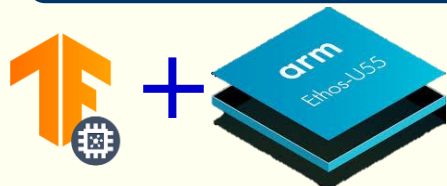
Model Optimization (Quantization) to int8 tflite



TensorFlow Lite
(int8 .tflite format)

Use python tflite interpreter
to check int8 tflite inference result

Pass Vela Compiler



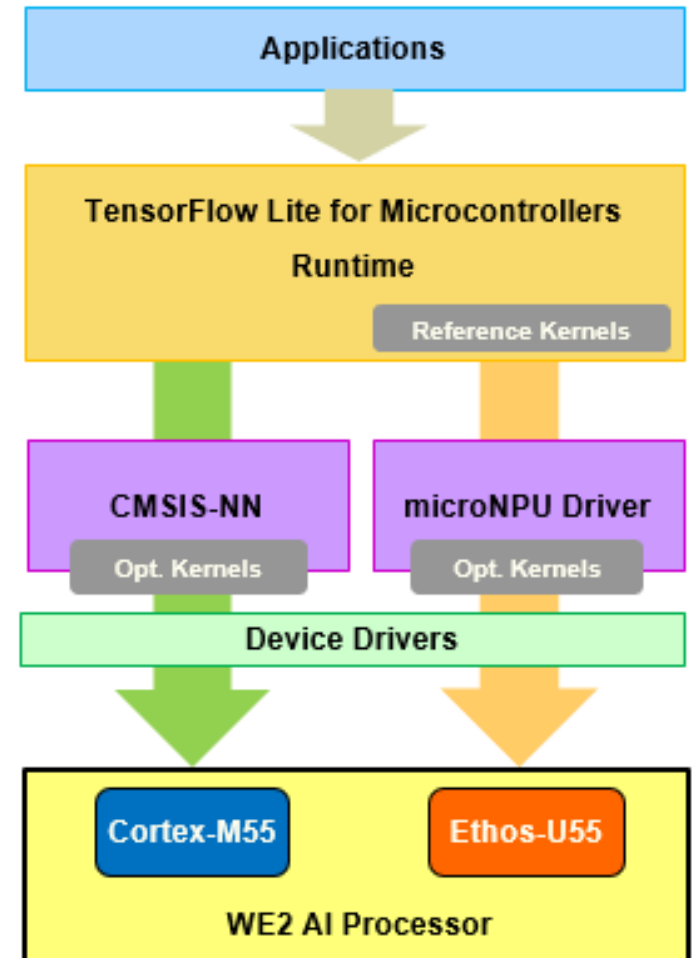
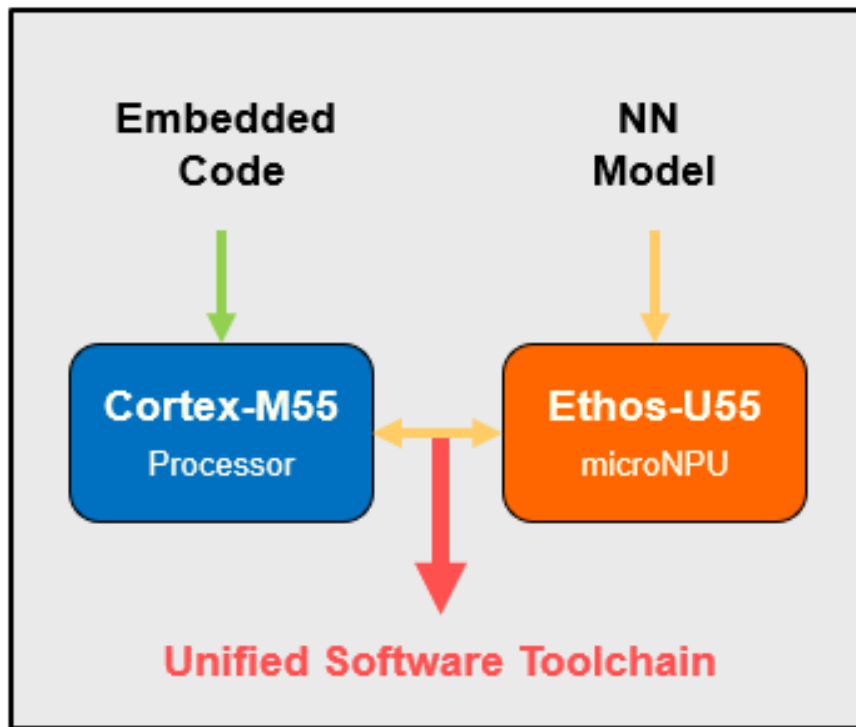
int8 vela model
(int8 .tflite format with ethosu operator)

Use FVP (Fixed Virtual Platforms)
to check int8 vela tflite inference result

Build firmware(FW) &
Flash FW and vela model to WE2

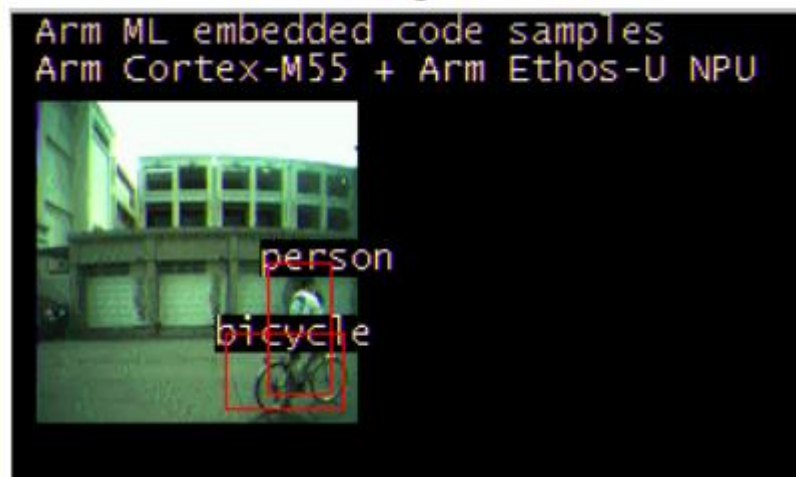
Open-Source Unified Software Toolchain

- An open-source unified toolchain for Ethos-U55 and Cortex-M55 simplifies developer creation of intelligence applications.



Fixed Virtual Platforms (FVP)

- Running at speeds comparable to the real hardware, Fixed Virtual Platforms (FVPs) are complete simulations of an Arm system, including processor, memory and peripherals.
- We provide the examples about how to run the model at FVP on Himax github.
 - ❖ https://github.com/HimaxWiseEyePlus/ML_FVP_EVALUATION
 - ❖ https://github.com/HimaxWiseEyePlus/YOLOv8_on_WE2?tab=readme-ov-file#deploy-yolov8n-to-fvp



Build Firmware

- WE2 Firmware Development Flow

Use Himax SDK API for C or C++ to program

make tool & Arm GNU Toolchain

Use Himax image gen tool to generate img file

Program Code

Compile Project

Create .elf file

Generate .img file

[https://github.com/HimaxWiseEyePlus/Seeed Grove Vision AI Module V2?tab=readme-ov-file#build-the-firmware-at-linux-environment](https://github.com/HimaxWiseEyePlus/Seeed-Grove-Vision-AI-Module-V2?tab=readme-ov-file#build-the-firmware-at-linux-environment)

Program Code



Third-party library

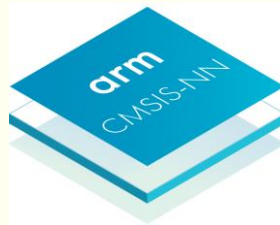
arm CMSIS-CV

arm CMSIS-DSP

M-profile
Vector
Extension
(MVE)



Himax SDK

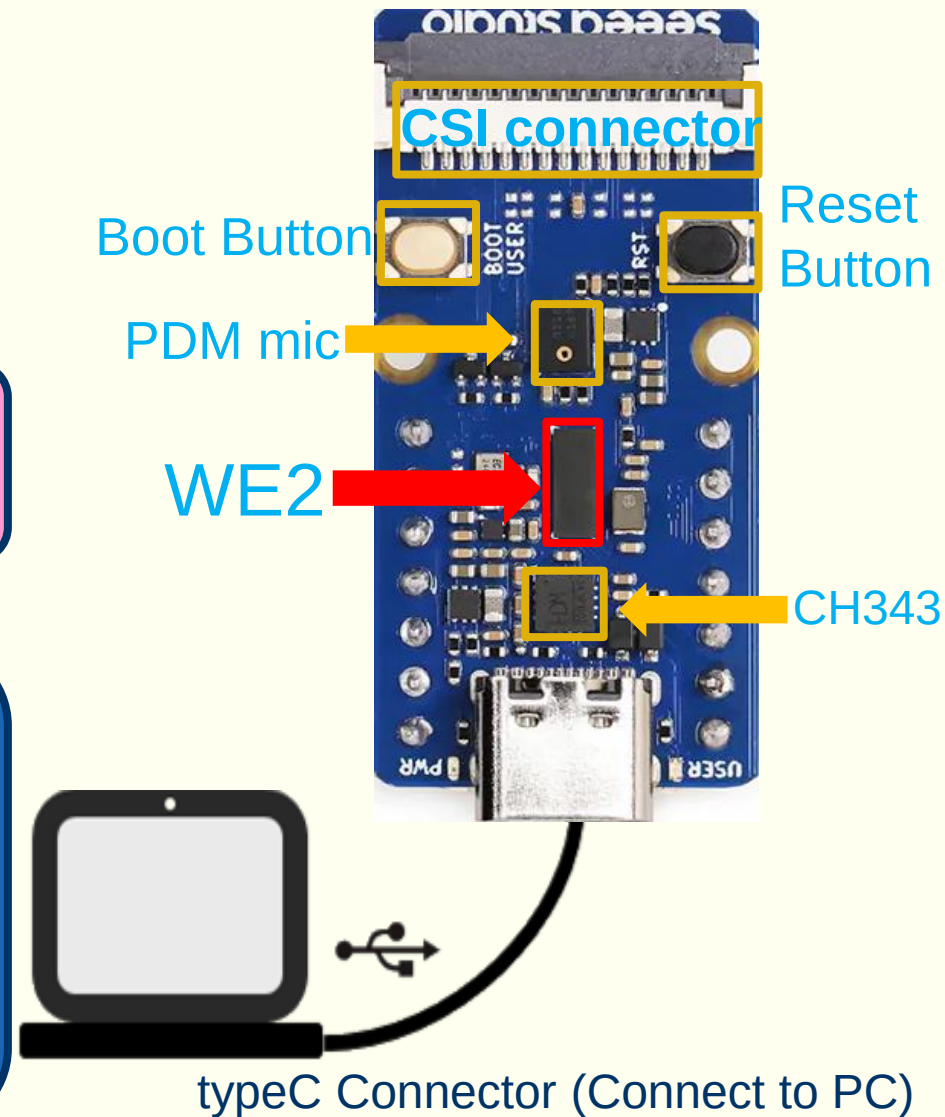
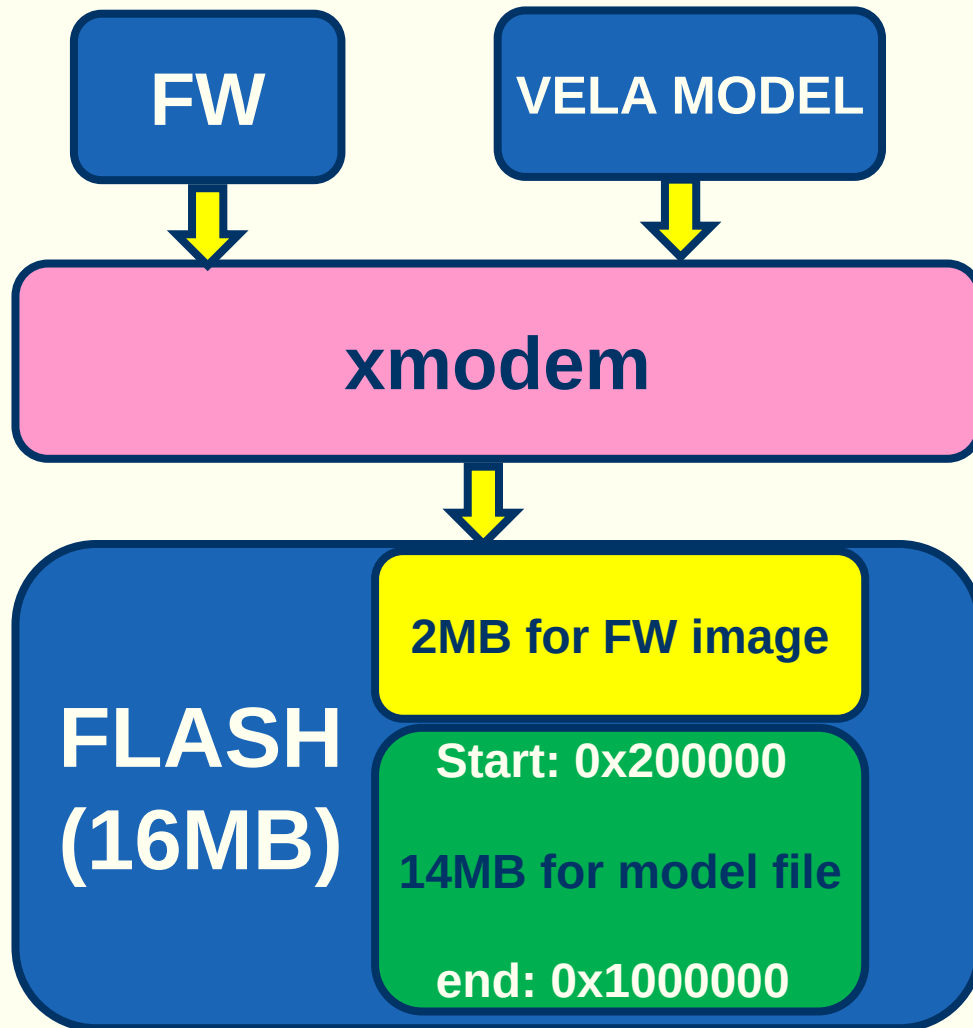


+
TensorFlow Lite
for microcontroller



Flash FW and vela model to Grove Vision AI V2(WE2)

FW should be not
bigger than 1MB



AI Model Optimization (MOT)



AI Model Optimization (MOT)

- Tensorflow Keras Yolo-Fastest/Yolo-Fastest XL with Model Optimization

- https://github.com/HimaxWiseEyePlus/Model_Training_Example/tree/main/Yolo_Fastest_series_with_MOT_TF_Keras

- Yolo-Fastest

Model	Yolo-Fastest	Yolo-Fastest	Yolo-Fastest MOT	Yolo-Fastest	Yolo-Fastest	Yolo-Fastest MOT
input size	RGB 320x320x3	RGB 320x320x3	RGB 320x320x3	RGB 256x256x3	RGB 256x256x3	RGB 256x256x3
MOT	X	X	O	X	X	O
COCO AP50	23.9	23.9	22.5	20.9	20.9	19.6
vela version	3.9.0	3.9.0	3.9.0	3.9.0	3.9.0	3.9.0
Optimise SRAM Size by Vela	X	O	O	X	O	O
Total SRAM(KB)	1200.91	820.00	820.00	768.91	528.00	528.00
Total off-chip Flash(KB)	505.31	548.70	429.70	511.52	545.64	432.28

AI Model Optimization (MOT)

- Quantization

Post-training quantization

Pre-trained model

Calibration data

Calibration

Quantization

Quantized model

Quantization-aware training

Pre-trained model

Quantization

Training data

Retraining/Finetuning

Quantized model

AI Model Optimization (MOT)

- Quantization

- ❖ You can open the tflite model file by [Netron.app](https://netron.app)
- ❖ You can check the tflite input and output which should be int8 format after Quantization.



MODEL PROPERTIES

format: TensorFlow Lite v3

description: TOCO Converted.

INPUTS

input	name: input	tensor	quantization: linear	location: 88
		int8[1,96,96,1]	$-1 \leq 0.007843137718737125 * (q + 1) \leq 1$	

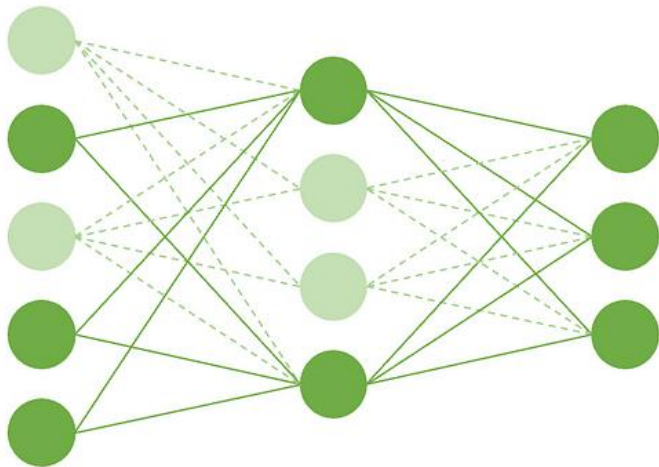
OUTPUTS

MobilenetV1/Pr...	name: MobilenetV1/Predictions/Reshape_1	tensor	quantization: linear	location: 87
		int8[1,2]	$0.00390625 * (q + 128)$	

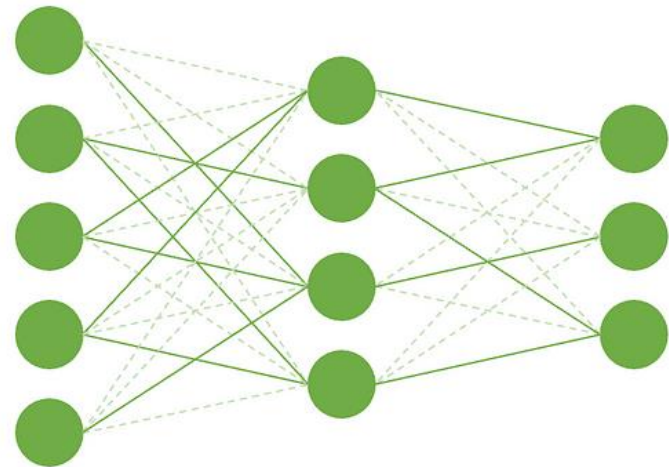
AI Model Optimization (MOT)

- Pruning

Structured Pruning

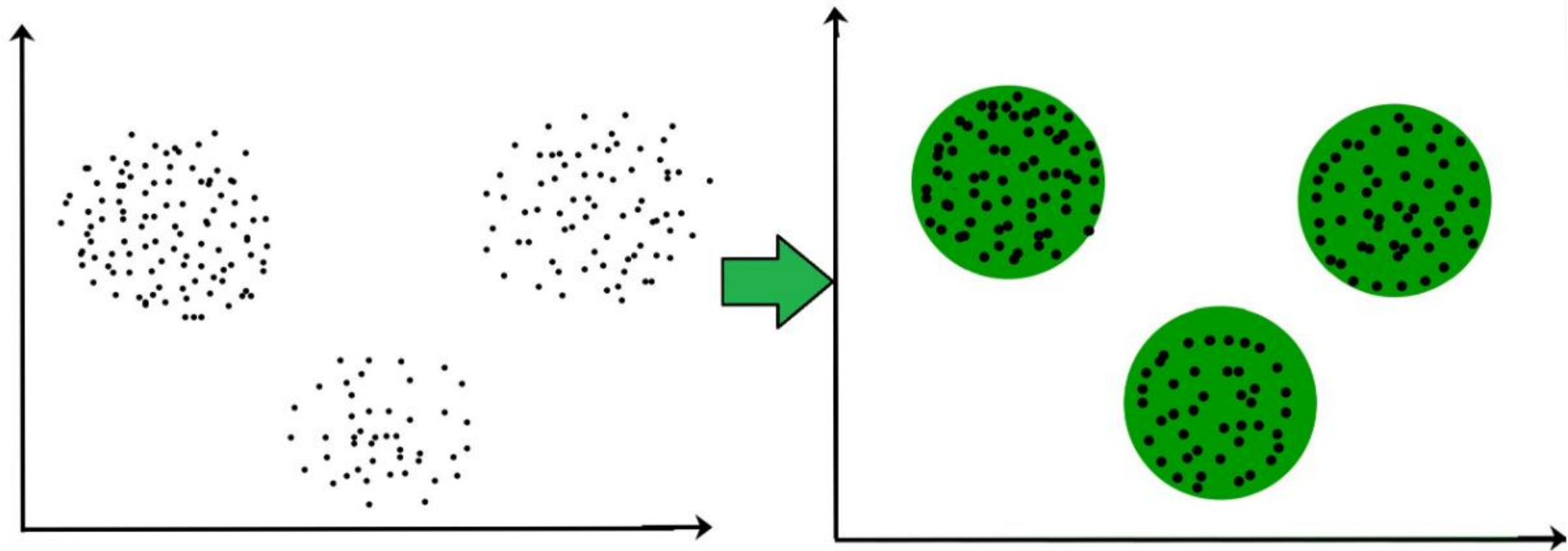


Unstructured Pruning



AI Model Optimization (MOT)

- Cluster



AI Model Optimization (MOT)

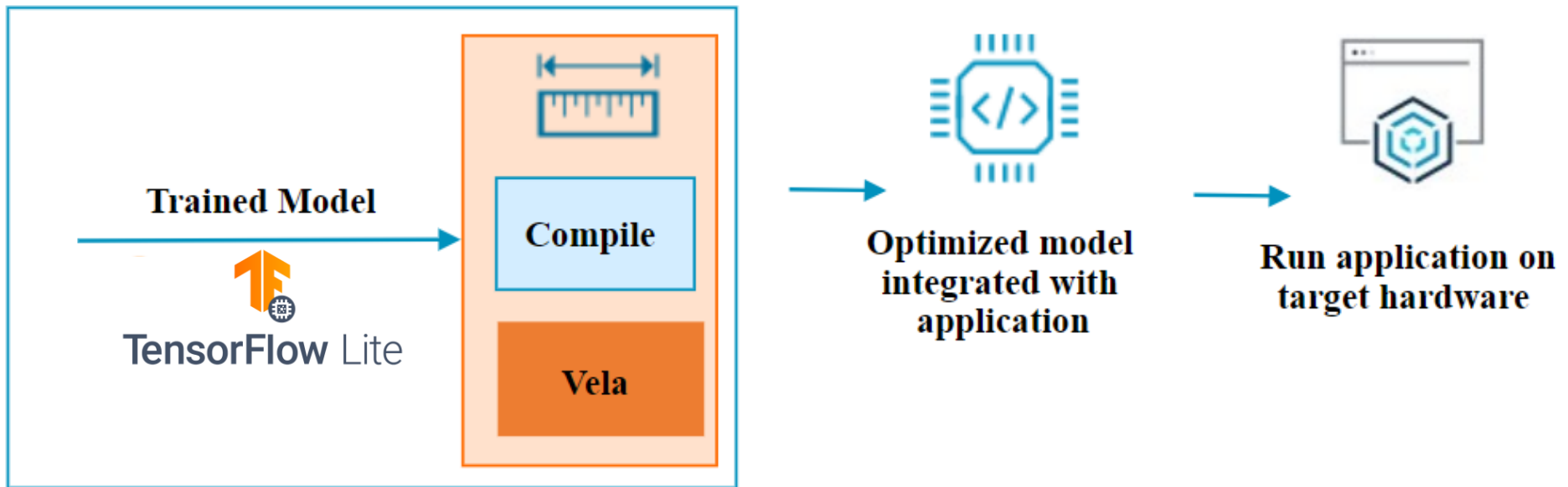
- After MOT and compress by vela, you will observe

- Yolo-Fastest

Model	Yolo-Fastest	Yolo-Fastest	Yolo-Fastest MOT	Yolo-Fastest	Yolo-Fastest	Yolo-Fastest MOT
input size	RGB 320x320x3	RGB 320x320x3	RGB 320x320x3	RGB 256x256x3	RGB 256x256x3	RGB 256x256x3
MOT	X	X	O	X	X	O
COCO AP50	23.9	23.9	22.5	20.9	20.9	19.6
vela version	3.9.0	3.9.0	3.9.0	3.9.0	3.9.0	3.9.0
Optimise SRAM Size by Vela	X	O	O	X	O	O
Total SRAM(KB)	1200.91	820.00	820.00	768.91	528.00	528.00
Total Off-chip Flash(KB)	505.31	548.70	429.70	511.52	545.64	432.28

AI Model Optimization

- Ethos-u-Vela
 - ❖ Ethos-U Vela development flow



Generate an optimized model for Ethos -U

AI Model Optimization

- Ethos-u-Vela
 - ❖ Ethos-U Vela development flow

How to use HIMAX config file to generate vela model

- Go under vela folder

```
cd vela
```



- Install necessary package:

```
pip install ethos-u-vela
```

himax_vela.ini file can be
download at
https://github.com/HimaxWiseEyePlus/ML_FVP_EVALUATION/blob/main/vela/himax_vela.ini



- Run vela with himax config ini file with mac=64 and the person detect example tflite model

```
vela --accelerator-config ethos-u55-64 --config himax_vela.
```



https://github.com/HimaxWiseEyePlus/ML_FVP_EVALUATION?tab=readme-ov-file#how-to-use-himax-config-file-to-generate-vela-model

AI Model Optimization

● Ethos-u-Vela

❖ Ethos-U Vela development flow

- You will see the vela report on the terminal: (Notes: Total SRAM used less than 900 KB will be better)

```
kris@kris-HP-ProBook-430-G7:~/Desktop/FVP_upload_to_github/ML_FVP_EVALUATION/vela$ vela --accelerator-config ethos-u55-64 --config himax_vela.ini --system-config My_Sys_Cfg --memory-mode My_Mem_Mode_Parent --output-dir ./img_person_detect ./img_person_detect/person_int8_model.tflite
```

Network summary for person_int8_model

Accelerator configuration	Ethos_U55_64
System configuration	My_Sys_Cfg
Memory mode	My_Mem_Mode_Parent
Accelerator clock	400 MHz
Design peak SRAM bandwidth	3.20 GB/s
Design peak Off-chip Flash bandwidth	0.05 GB/s
Total SRAM used	72.72 KiB
Total Off-chip Flash used	266.19 KiB

63 passes fused into 2
0/132 (0.0%) operations falling back to the CPU

Average SRAM bandwidth	0.26 GB/s
Input SRAM bandwidth	0.39 MB/batch
Weight SRAM bandwidth	0.61 MB/batch
Output SRAM bandwidth	0.27 MB/batch
Total SRAM bandwidth	1.29 MB/batch
Total SRAM bandwidth per input	1.29 MB/inference (batch size 1)
Average Off-chip Flash bandwidth	0.05 GB/s
Input Off-chip Flash bandwidth	0.00 MB/batch
Weight Off-chip Flash bandwidth	0.23 MB/batch
Output Off-chip Flash bandwidth	0.00 MB/batch
Total Off-chip Flash bandwidth	0.23 MB/batch
Total Off-chip Flash bandwidth per input	0.23 MB/inference (batch size 1)

Neural network macs 7215492 MACs/batch
Network Tops/s 0.00 Tops/s

NPU cycles 1997673 cycles/batch
SRAM Access cycles 86048 cycles/batch
DRAM Access cycles 0 cycles/batch
On-chip Flash Access cycles 0 cycles/batch
Off-chip Flash Access cycles 11552 cycles/batch
Total cycles 2009130 cycles/batch

Batch Inference time 5.02 ms, 199.09 inferences/s (batch size 1)

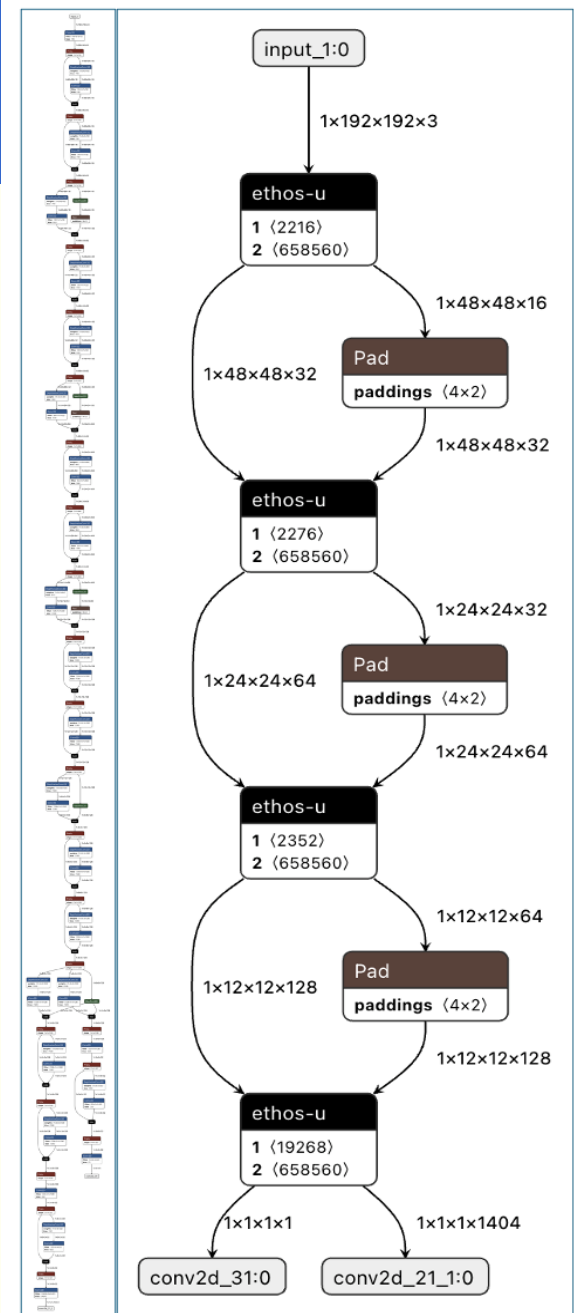
Check tensor arena size
can fit SRAM or not at
vela report (less than 900
KB will be better)

AI Model Optimization

- Ethos-u-Vela

- ❖ (Left) Graph of 8-bit quantized TensorFlow Lite model
- ❖ (Right) Graph of 8-bit quantized TensorFlow Lite model compiled with vela

You can open the tflite model file by [Netron.app](https://netron.app)



AI Model Optimization

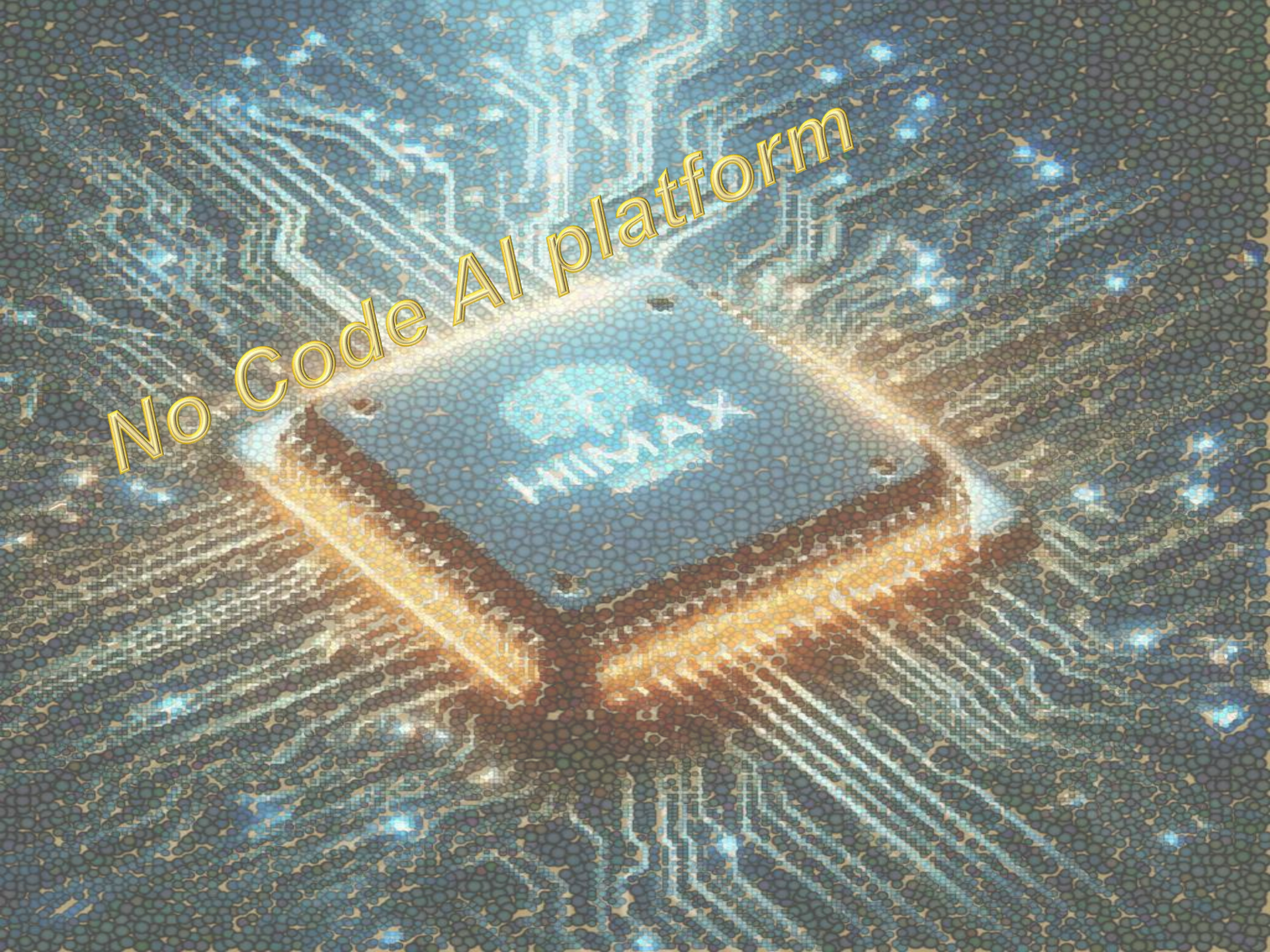
- Compare Model performance on M55 VS M55+U55

➤ MediaPipe Face Mesh model

	Cortex-M55 (without Helium)	Cortex-M55 + Ethos-U55
RAM (TFLM tensor arena size)	482 KB	434 KB
Flash (to store TFLite model)	748 KB	688 KB
Inference Latency (excludes pre and post processing)	4,436 ms	43 ms

➤ YOLOv8n pose model

	Cortex-M55 (without Helium)	Cortex-M55 + Ethos-U55
RAM (TFLM tensor arena size)	853 KB	551 KB
Flash (to store TFLite model)	3.23 MB	2.19 MB
Inference Latency (excludes pre and post processing)	61,486 ms	93 ms



No Code AI platform

No Code AI platform

- Seeed SenseCraft AI

- ❖ Local Device - SenseCraft AI (seed.cc)

SenseCraft | SenseCraft AI 模型列表 设备工作空间 关于SenseCraft AI [] [A] [C] [R] kris

根据模型名搜索 Q

369 模型 清空

Grove - Vision AI V2 x

模型任务

目标检测 分类检测 图像分割 姿态检测

生成式模型

支持的设备类型

reComputer Jetson XIAO ESP32S3 Sense

Grove - Vision AI V2 SenseCAP Watcher

发布者

SenseCraft AI Abhinav krishna N Hendra

公共AI模型 我的模型

船舶检测



一种旨在识别船舶的 AI 模型，可部署在 Seeed Studio Grove Vision AI (V2) 设备上，使用 Yolo-Word 算法...

目标检测 Grove - Vision AI V2

SenseCraft AI

篮球检测



该 AI 模型由 Yolo-Word 算法生成，专为 Seeed Studio Grove Vision AI (V2) 设备设计，能够有效识别和检测...

目标检测 Grove - Vision AI V2

SenseCraft AI

婴儿车检测



该 AI 模型由 Yolo-Word 算法生成，专为 Seeed Studio Grove Vision AI (V2) 设备设计，能够高效识别和检测...

目标检测 Grove - Vision AI V2

SenseCraft AI

钢笔/铅笔检测



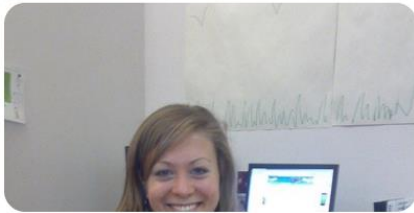
利用 Yolo-Word 算法，该 AI 模型专为 Seeed Studio Grove Vision AI (V2) 设备而设计，可有效检测和识别...

垃圾桶检测



该 AI 模型由 Yolo-Word 算法生成，专为 Seeed Studio Grove Vision AI (V2) 设备设计，能够高效识别和检测...

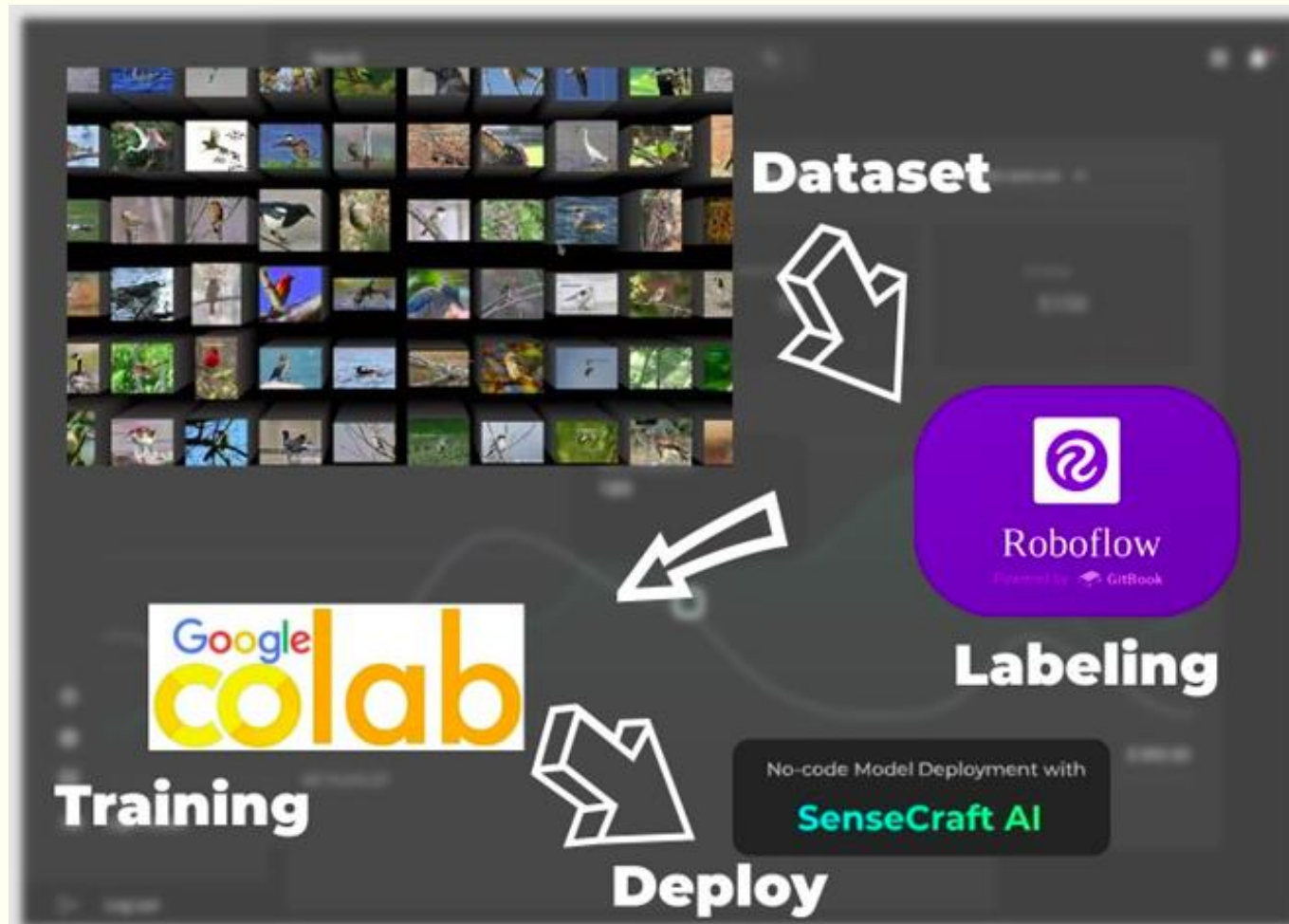
环检测



该 AI 模型由 Yolo-Word 算法生成，专为 Seeed Studio Grove Vision AI (V2) 设备设计，能够有效识别和检测...

No Code AI platform

- Seeed SenseCraft AI
 - ❖ [Quick Start | Seeed Studio Wiki](#)



No Code AI platform

- Edge Impulse

- ❖ <https://docs.edgeimpulse.com/docs/edge-ai-hardware/mcu+-ai-accelerators/himax-seeed-grove-vision-ai-module-v2-wise-eye-2>

The screenshot displays the Edge Impulse web interface. On the left is a sidebar with navigation links: Dashboard, Devices, Data acquisition, Experiments, and Impulse design (which is expanded to show 'Create impulse', 'Image', 'Object detection', 'Retrain model', 'Live classification', 'Model testing', and 'Deployment'). The main workspace is titled 'Impulse #1' and contains a descriptive text box: 'An impulse takes raw data, uses signal processing to extract features, and then uses a learning block to classify new data.' Below this, four colored blocks represent different stages of the pipeline: 1. 'Image data' (red block) with input axes for 'Image' (96x96) and a 'Resize mode' dropdown set to 'Fit'. 2. 'Image' (light blue block) with a name 'Image' and an input axis 'Image'. 3. 'Object Detection (Images)' (purple block) with a name 'Object detection', input features including 'Image' (checked), and output features '1 (person)'. 4. 'Output features' (teal block) showing '1 (person)' and a 'Save impulse' button. A help icon is visible in the bottom right corner of the workspace.

No Code AI platform

● Edge Impulse



- Dashboard
- Devices
- Data acquisition
- Experiments
- Impulse design
 - Create impulse
 - Image
 - Object detection
 - Retrain model
 - Live classification
 - Model testing
 - Deployment
- Versioning
- GETTING STARTED
 - Documentation
 - Forums

Himax / testtt ENTERPRISE

Target: Himax WiseEye2 ...

Dataset

Data sources

Synthetic data

Labeling queue (1)

Auto-labeler

DATA COLLECTED

333 items

TRAIN / TEST SPLIT

80% / 20%

Collect data

Connect a device to start building your dataset.

Dataset

Training (265) Test (68)

SAMPLE NAME	LABELS	ADDED
sample.53tt7tvb	person	7月 30日 2024年, 15:...
sample.53tl94kr	person	7月 30日 2024年, 13:...
Skype_Video.s985	person	7月 01日 2024年, 14:...
Skype_Video.s982	person, person, pers...	7月 01日 2024年, 14:...
Skype_Video.s979	person, person	7月 01日 2024年, 14:...
Skype_Video.s976	person, person	7月 01日 2024年, 14:...
Skype_Video.s970	person, person, pers...	7月 01日 2024年, 14:...
Skype_Video.s967	person, person, pers...	7月 01日 2024年, 14:...
Skype_Video.s964	person, person, pers...	7月 01日 2024年, 14:...
Skype_Video.s961	person, person, pers...	7月 01日 2024年, 14:...
Skype_Video.s955	person, person, pers...	7月 01日 2024年, 14:...
Skype_Video.s952	person, person, pers...	7月 01日 2024年, 14:...

RAW DATA

sample.53tt7tvb

Metadata

No metadata.

No Code AI platform

- Ultralytics Hub (Yolov8)

- ❖ <https://hub.ultralytics.com/>

- ❖ https://github.com/HimaxWiseEyePlus/YOLOv8_on_WE2

The screenshot displays the Ultralytics Hub (Yolov8) interface. On the left is a sidebar with navigation links: Home, Datasets, Projects, Models, Integrations, Support, and Trash. The main content area is titled 'Home' and features a 'Get Started' section with a 'Start Welcome Tutorial' button. Below this are three main sections: 'Datasets' (with an 'Upload Dataset' button), 'Projects' (with a 'Create Project' button), and 'Models' (with a 'Train Model' button). On the right side, there is a user profile section for 'kris' with a 'Hello kris!' greeting, storage usage (702.8 MB / 20.0 GB), current plan (FREE), and an 'Upgrade' button. At the bottom, there is a 'Download the Ultralytics App' button and a mobile app preview.

Install tool at Windows environment

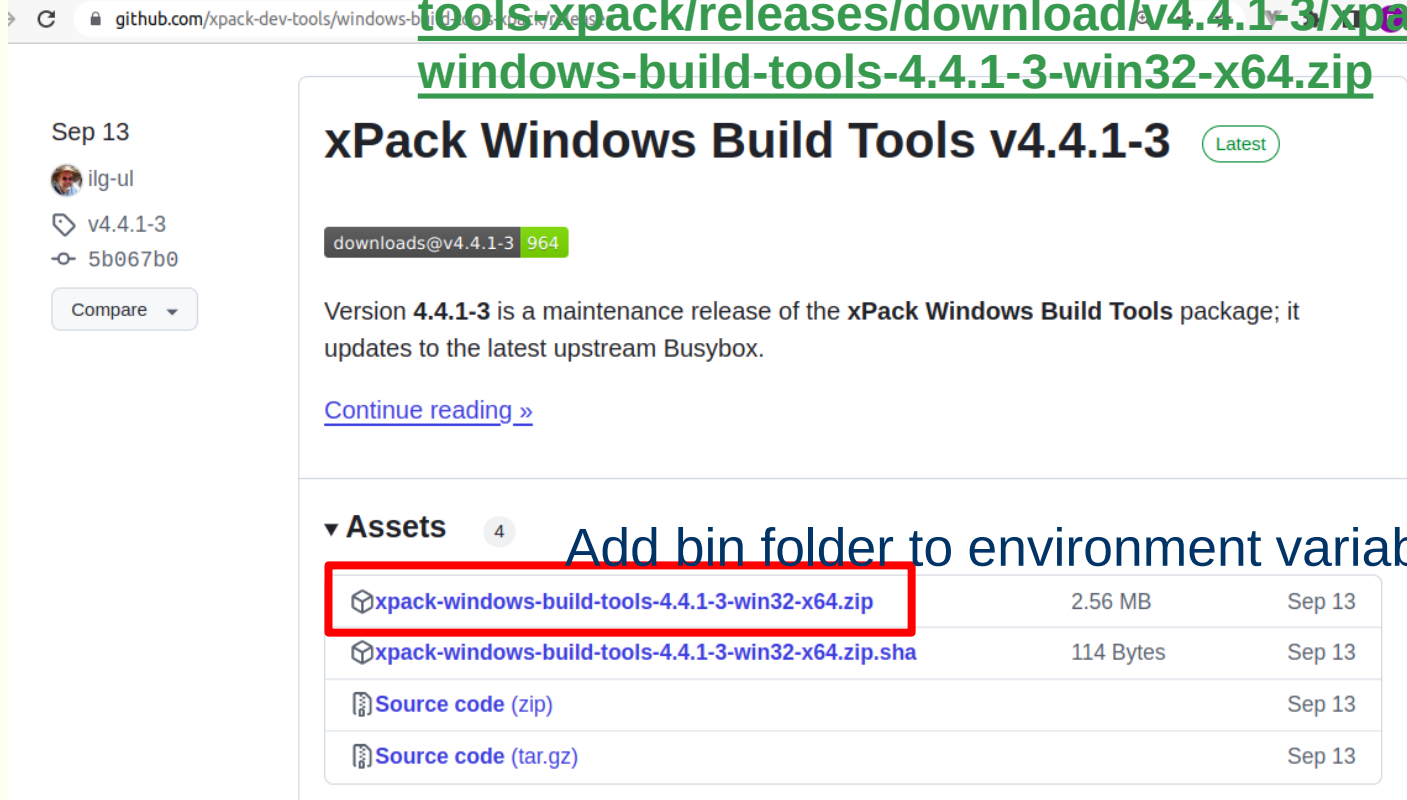
- Download the source code from himax github

The screenshot shows the GitHub repository page for **HimaxWiseEyePlus / Seeed_Grove_Vision_AI_Module_V2**. The repository is public and has 84 stars, 36 forks, and 6 watchers. The 'Code' button is highlighted with a red box. A dropdown menu is open, showing the 'Clone' option, which is also highlighted with a red box. The 'Download ZIP' option is also highlighted with a red box. The URL `https://github.com/HimaxWiseEyePlus/Seeed_Grove_Vision_AI_Module_V2` is highlighted with a red box. The text **Git clone** is overlaid on the image.

Install tool at Windows environment

- Step 1: Install the **make** command for prerequisites , you can reference

<https://github.com/xpack-dev-tools/windows-build-tools-xpack/releases/download/v4.4.1-3/xpack-windows-build-tools-4.4.1-3-win32-x64.zip>



Sep 13
ilg-ul
v4.4.1-3
5b067b0
Compare

xPack Windows Build Tools v4.4.1-3 Latest

downloads@v4.4.1-3 964

Version **4.4.1-3** is a maintenance release of the **xPack Windows Build Tools** package; it updates to the latest upstream Busybox.

[Continue reading »](#)

▼ Assets 4

xpack-windows-build-tools-4.4.1-3-win32-x64.zip	2.56 MB	Sep 13
xpack-windows-build-tools-4.4.1-3-win32-x64.zip.sha	114 Bytes	Sep 13
Source code (zip)		Sep 13
Source code (tar.gz)		Sep 13

Add bin folder to environment variable

- Step 1.1 : Install **CH343 UART** driver ([CH343SER.exe](#))

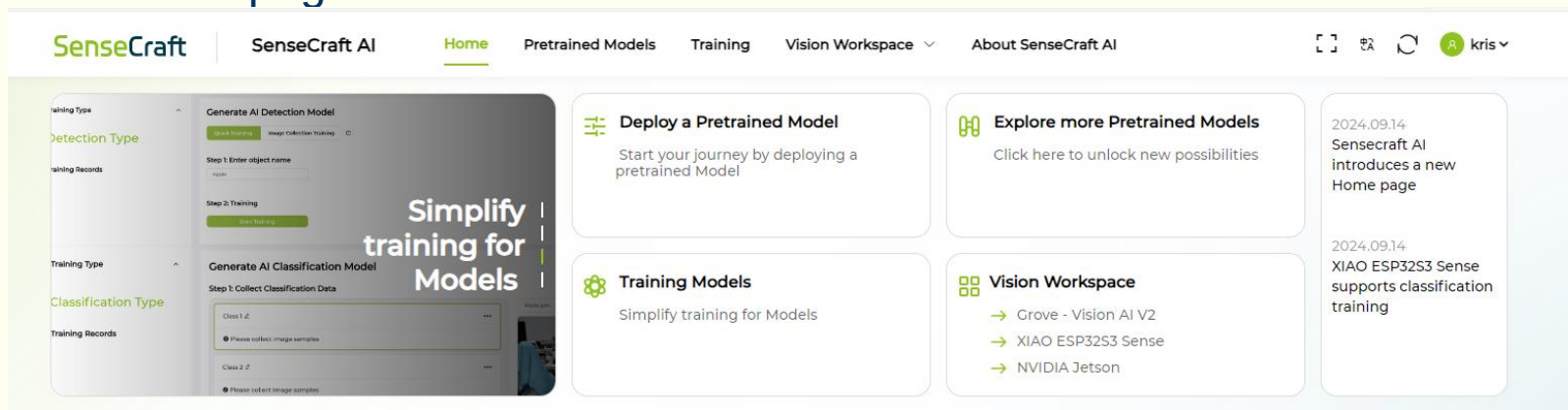
Install tool at Windows environment

- Step 2: Download Arm GNU Toolchain
 - ❖ [arm-gnu-toolchain-13.2.rel1-mingw-w64-i686-arm-none-eabi.zip](#)
- Step 3: Extract the file
 - ❖ `tar -xvf arm-gnu-toolchain-13.2.rel1-mingw-w64-i686-arm-none-eabi.zip`
- Step 4: Add arm-gnu-toolchain-13.2.rel1-mingw-w64-i686-arm-none-eabi/bin/ to PATH
 - ❖ `setx PATH "%PATH%;[location of your gnu-toolchain-13.2 ROOT]\arm-gnu-toolchain-13.2.rel1-mingw-w64-i686-arm-none-eabi\bin"`

Register Seeed SenseCraft AI

- Seeed SenseCraft AI

- ❖ Home page



Start your journey: Deploy a pretrained Model

Step 1: Discover Pretrained Models

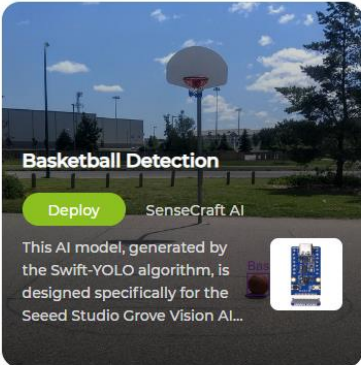
Grove - Vision AI V2

XIAO ESP32S3 Sense

Basketball Detection

Deploy SenseCraft AI

This AI model, generated by the Swift-YOLO algorithm, is designed specifically for the Seeed Studio Grove Vision AI...



[Explore more Pretrained Models →](#)[Training →](#)



Drive for better vision